



Fortinet Cybersecurity Academy: Configuring a FortAP With WPA2-PSK and WPA2- Enterprise Local Auth

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Purpose

The purpose of this lab is to learn about the FortiAP and how to connect it to the FortiGate firewall to set up two different types of WLANs to access the internet.

Background

Fortinet is only of the world's largest cybersecurity companies. They design thier own custom ASICs, Operating System (FortiOS), and build everything into a unified platform called the Fortinet Scurity fabric. They are known for their flagshipproduct, the FortiGate firewall, as well as a wide variety of other networking and security devices, such as the FortiSwitch and the FortiAP, and services such as FortiSandbox, FortiAuthenticator, FortiVoice, and FortiDDoS. Fortigate is an S&P 500 component and is listed on the NASDAQ as \$FTNT.

The FortiAP 421E is an access point developed by Fortinet. It is designed for Small Office Home Office Networks as well as remote branch locations that require enterprise-grade security in a small form factor. It runs on Fortinet's proprietary FortiOS operating system and is powered by Fortinet's SP5 security processing unit, which provides hardware acceleration for firewall and VPN functions, allowing it to deliver up to 5 Gbps of firewall throughput. It supports a variety of authentication methods, including WPA and WPA2 with either802.1X or PSK, WEP, and a MAC blacklist/whitelist. It can form a mesh with other. FortiAP devices to ensure more uniform connectivity across a site. The 421E can also handle a maximum of 512 clients per radio split across a maximum of 16 SSIDs.

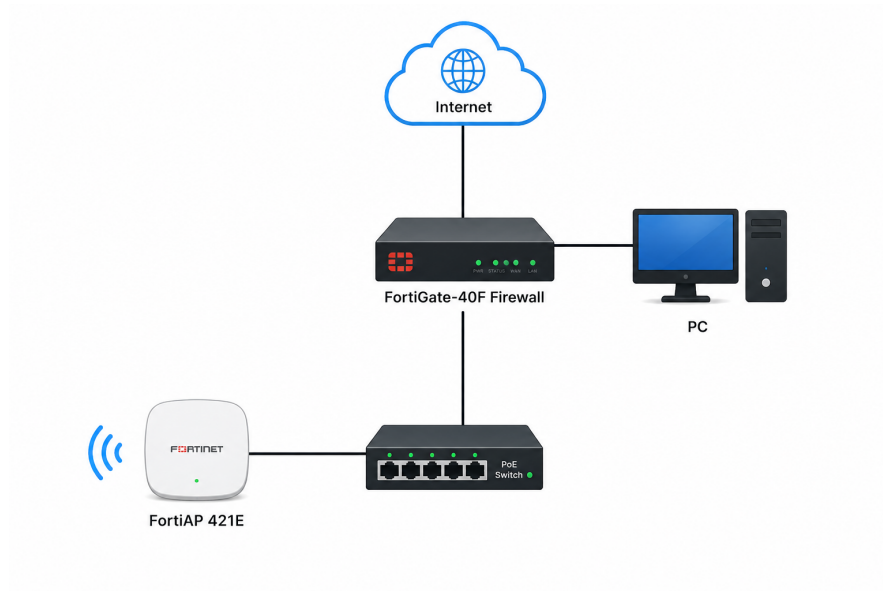
Wi-Fi Protected Access, or WPA, is a wireless network security standard created by the Wi-Fi Alliance to protect data transmitted over Wi-Fi networks. WPA was introduced in 2003 as a direct replacement for the aging and critically flawed WEP (Wired Equivalent Privacy) protocol, which had been shown to be easily broken by attackers using widely available tools. There are three versions of WPA, being named WPA, WPA2, and WPA3 respectively. The first generation of WPA was released in 2003, with the second version being released just a year later in 2004. In 2018, the third generation was released. WPA uses TKIP (Temporal Key Integrity Protocol), which dynamically generates a new 128-bit key for each packet, helping to prevent the replay attacks that plagued WEP. WPA2 replaced TKIP with the stronger CCMP (Counter Mode with Cipher Block Chaining Message Authentication Code Protocol), WPA3 retains support for CCMP but introduces GCMP (Galois/Counter Mode Protocol), a more efficient ane stronger encryption method. WPA3 also implements SAE (Simultaneous Authentication of Equals), replacing the older Pre-Shared Key handshake to better protect against offline dictionary attacks.

WPA can use two different methods of authentication: Personal and Enterprise. Personal authentication uses a pre-shared 256 bit key, meaning that all devices authenticate using the same password. Enterprise authentication uses a RADIUS (Remote Authentication Dial In User Service) server, meaning that each user authenticates using their own username and password.

Lab Summary

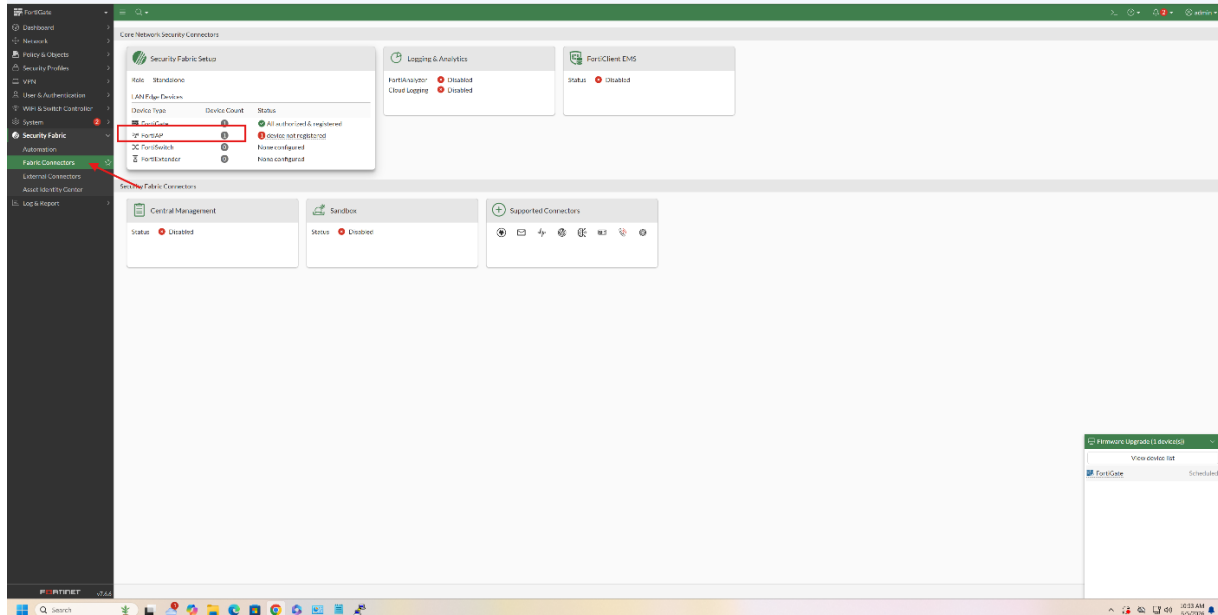
In this lab, we set up the FortiAP 42E with a Fortigate Firewall with two WLANs, running WPA2-Personal with PSK and WPA2-Enterprise with Local authentication.

Topology

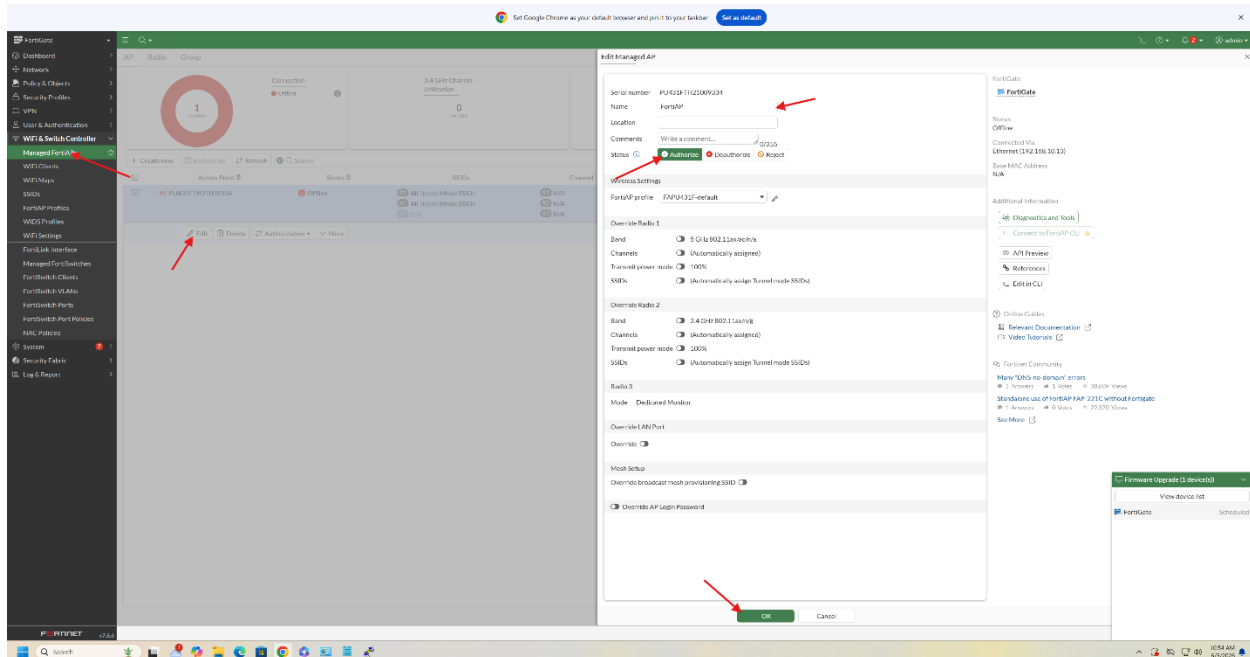


Lab Commands

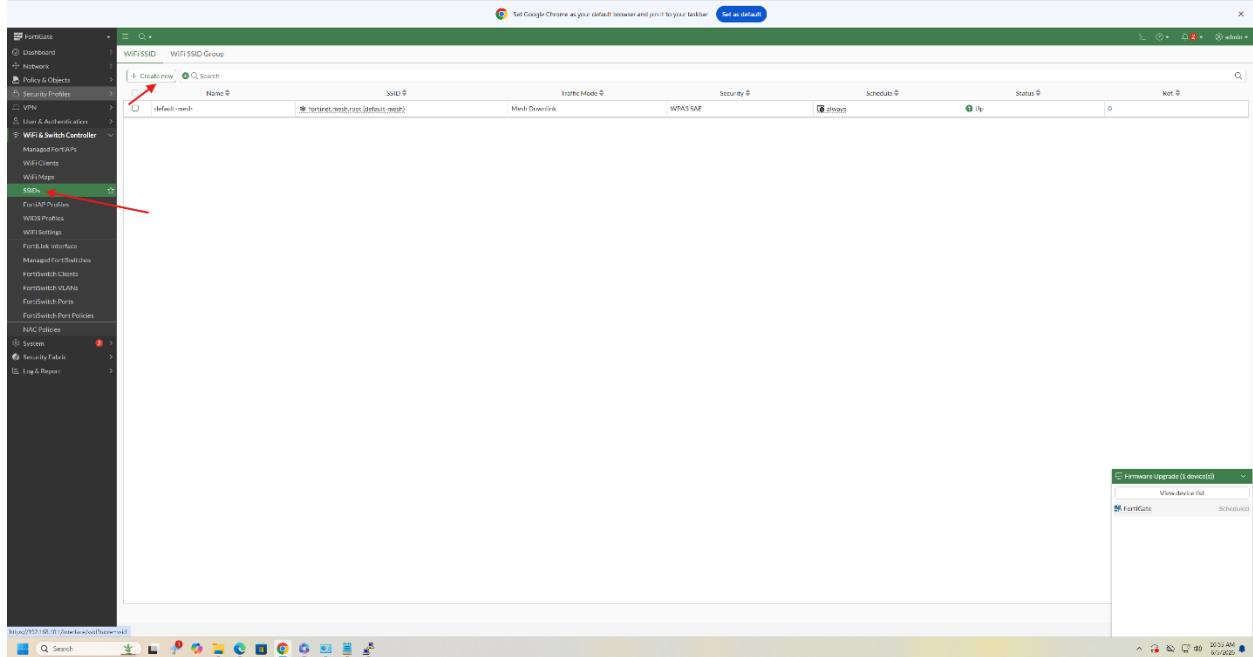
Once you connect the AP to the firewall go to the firewall to the Dashboard, and under Security Fabric, ensure that one (1) FortiAP device is visible.



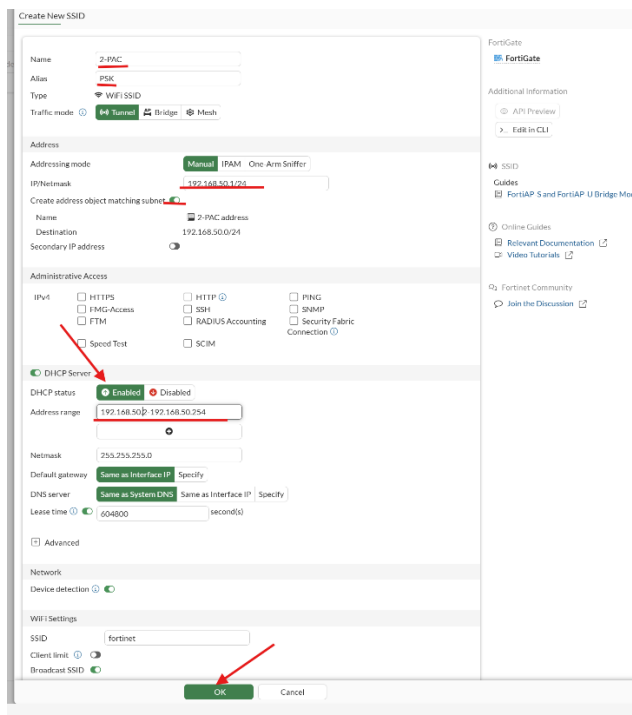
Go to WiFi & Switch Controller > Managed FortiAPs, click on the unauthorized AP, and click Authorization > Authorize.



Next, we will set up the WPA2-PSK SSID. Go to WiFi & Switch Controller > SSIDs, and click Create New



Configure an appropriate name. Give the SSID an appropriate IP and netmask, is enabled. The DHCP server settings should automatically match the network settings configured in the Address section.



Set the SSID to the same as the name, ensure that the SSID is being broadcasted, set the security mode to WPA2 Personal, and configure an appropriate passphrase. Then Click ok

Create New Policy

Schedule: ☒ always

Action: ☒ ACCEPT ☐ DENY

Incoming interface: ☒ 2-PAC (2-PAC)

Outgoing interface: ☒ Internet (wan)

Source & Destination

Source: ☒ 2-PAC address

User/group:

Destination: ☒ all

Service: ☒ ALL

Firewall/Network Options

NAT: ☒ Use Outgoing Interface Address ☐ Use Dynamic IP Pool

Source port translation: ☒ Always ☐ When port conflicts ☐ Never

Protocol options: ☒ PROT ☐ default

Security Profiles

AntiVirus: ☒

Web filter: ☒

DNS filter: ☒

Application control: ☒

IPS: ☒

SSL Inspection: ☒ SSL ☐ no-inspection

Logging Options

Log allowed traffic: ☒ Security events ☐ All sessions

Comments:

0/1023

Enable this policy: ☒

OK Cancel

Your PSK WLAN should now be fully set up. Confirm that it works by connecting another device, inputting the password configured earlier. As seen in the screenshot below, when connected to the 2-PAC network, my laptop is connected to the internet and can ping Google.com



```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

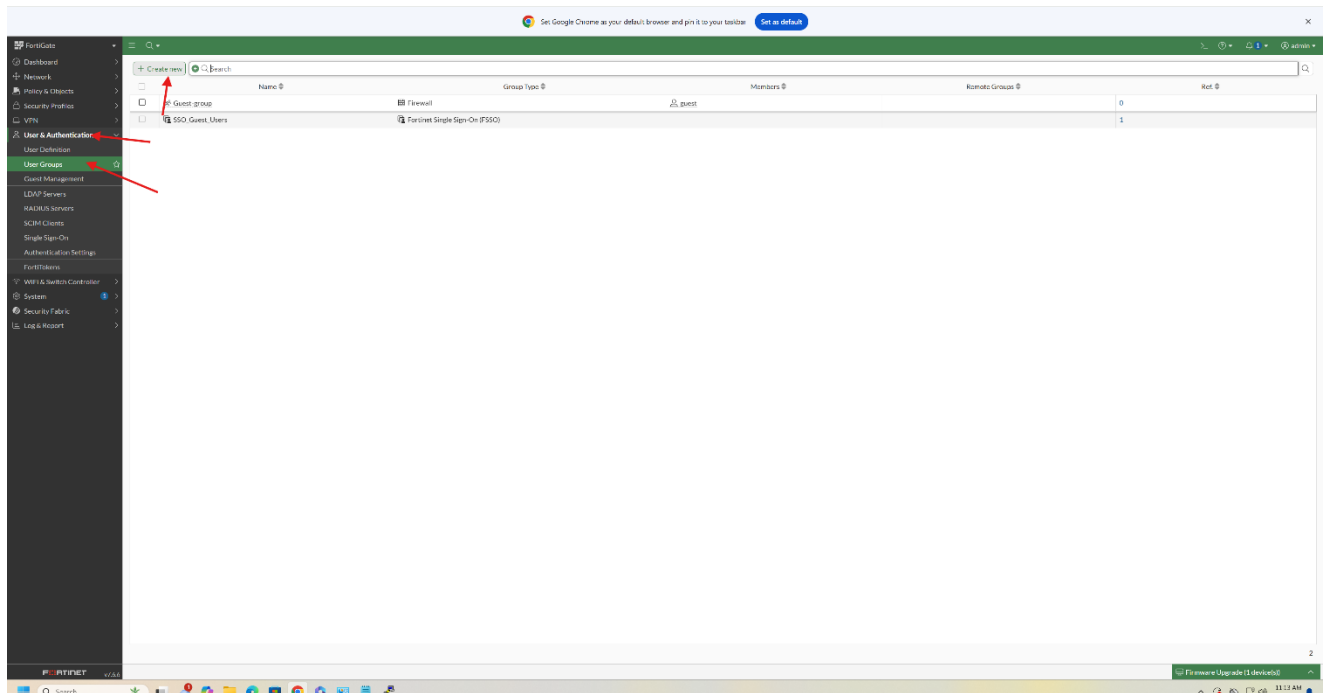
C:\Users\krishi>ping www.google.com

Pinging www.google.com [142.251.152.119] with 32 bytes of data:
Reply from 142.251.152.119: bytes=32 time=11ms TTL=112
Reply from 142.251.152.119: bytes=32 time=14ms TTL=112
Reply from 142.251.152.119: bytes=32 time=11ms TTL=112
Reply from 142.251.152.119: bytes=32 time=14ms TTL=112

Ping statistics for 142.251.152.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 14ms, Average = 12ms

C:\Users\krishi>
```

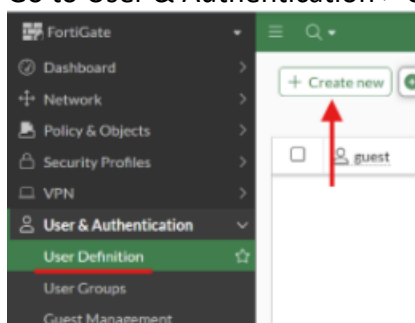
Next, you will configure Local User authentication for the WPA2 Enterprise WLAN. Go to User & Authentication > User Groups>create new



Choose a group name and click ok



Go to User & Authentication > User Definition and click Create new.



Set the User Type to Local User.

The screenshot shows the 'Users/Groups Creation Wizard' with the first step, 'User Type', selected. A list of user types is displayed: Local User, Remote RADIUS User, Remote TACACS+ User, Remote LDAP User, FSSO, FortiNAC User, and SAML User. The 'Local User' option is highlighted. At the bottom, there are three buttons: '< Back', 'Next', and 'Cancel'. A red arrow points to the 'Next' button.

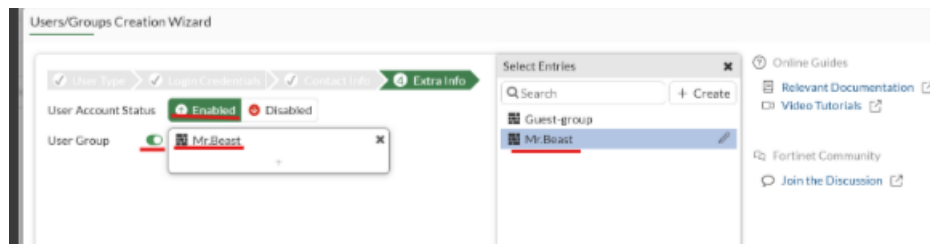
Set an appropriate username and password.

The screenshot shows the 'Users/Groups Creation Wizard' with the second step, 'Login Credentials', selected. The 'Username' field contains the text 'Local' and the 'Password' field contains a series of asterisks. Both fields have red error bars below them, indicating they are required or invalid. At the bottom, there are three buttons: '< Back', 'Next', and 'Cancel'.

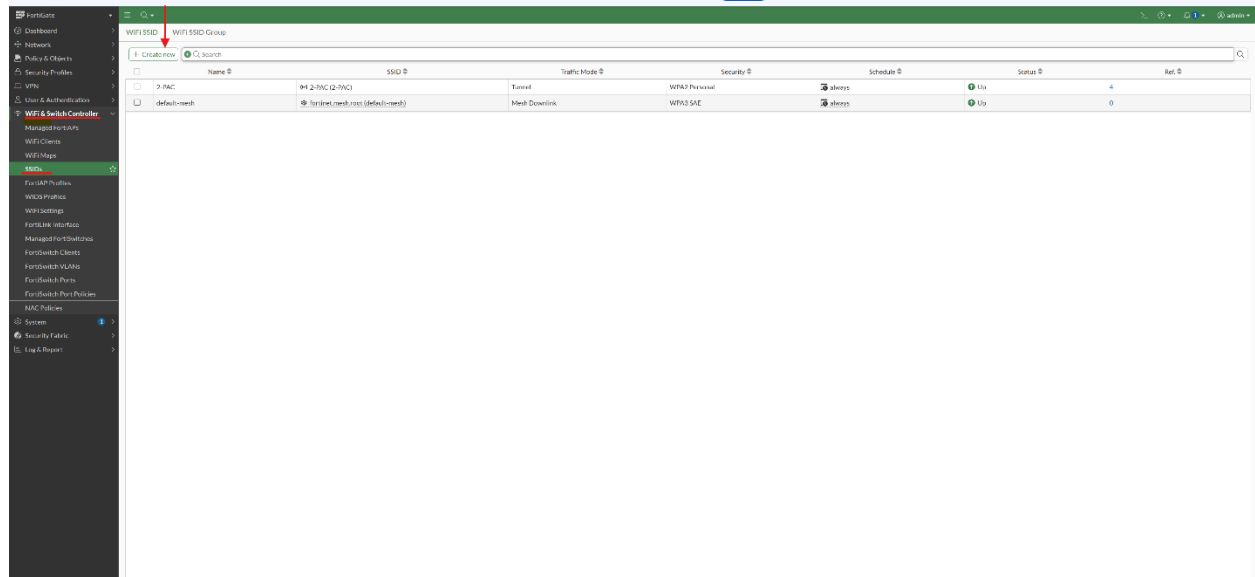
Under Contact Info, leave two-factor authentication off

The screenshot shows the 'Users/Groups Creation Wizard' with the third step, 'Contact Info', selected. The 'Two-factor Authentication' option is shown with a radio button that is not selected, indicating it is turned off.

Set the User Account Status to Enabled, enable User Group, click the plus icon, and select the User Group created earlier.



Next, we will set up the WPA2-Enterprise SSID. Go to WiFi & Switch Controller > SSIDs and click Create New > SSID.



Assign an appropriate name. Assign an appropriate IP/netmask. Ensure that a DHCP server for the subnet is created.

Create New SSID

Name: SaiKrish

Alias: enterprise

Type: WiFi SSID

Traffic mode: Tunnel Bridge Mesh

Address

Addressing mode: Manual IPAM One-Arm Sniffer

IP/Netmask: 192.168.94.1/24

Create address object matching subnet: ☒

Name: SaiKrish address

Destination: 192.168.94.0/24

Secondary IP address: ☐

Administrative Access

IPv4: ☐ HTTPS ☐ HTTP ☐ PING ☐ FMG-Access ☐ SSH ☐ SNMP ☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

☒ DHCP Server

DHCP status: Enabled Disabled

Address range: 192.168.94.2-192.168.94.254

Netmask: 255.255.255.0

Default gateway: Same as Interface IP Specify

DNS server: Same as System DNS Same as Interface IP Specify

Lease time: 604800 second(s)

☐ Advanced

FortiGate

FortiGate

Additional Information

API Preview

Edit in CLI

SSID

Guides

FortiAP-S and FortiAP-U Bridge Mode Security Profiles

Online Guides

Relevant Documentation

Video Tutorials

Fortinet Community

Join the Discussion

Set the SSID to the same as the name, ensure the SSID is broadcast, set the security mode to WPA2 Enterprise, set the authentication to Local, and add the user group created earlier.

Create New SSID

Network

Device detection: ☒

WiFi Settings

SSID: SaiKrish

Client limit: ☐

Broadcast SSID: ☒

Beacon advertising: ☐ Name ☐ Model ☐ Serial number

Security Mode Settings

Security mode: WPA2 Enterprise

Authentication: Local RADIUS Server

Int Boost

Client MAC Address Filtering

RADIUS server: ☐

Address group policy: Disable Allow Deny

Additional Settings

Schedule: always

Block Intra-SSID traffic: ☐

Optional VLAN ID: 0

Broadcast suppression: ☒ ARPs for known clients ☒ DI ICP unicast ☒ DHCP uplink ☒

Quarantine host: ☐

VLAN pooling: ☐

NAC profile: ☐

Traffic Shaping

Outbound shaping profile: ☐

Outbound bandwidth: ☐

Miscellaneous

Comments: 0/255

Status: Enabled Disabled

OK Cancel

FortiGate

FortiGate

Additional Information

API Preview

Edit in CLI

SSID

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Online Guides

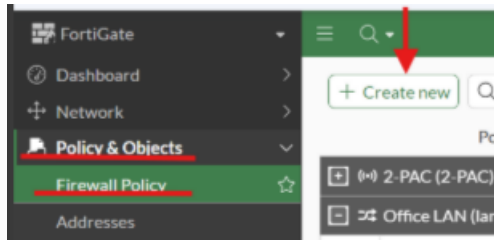
Relevant Documentation

Video Tutorials

Fortinet Community

Join the Discussion

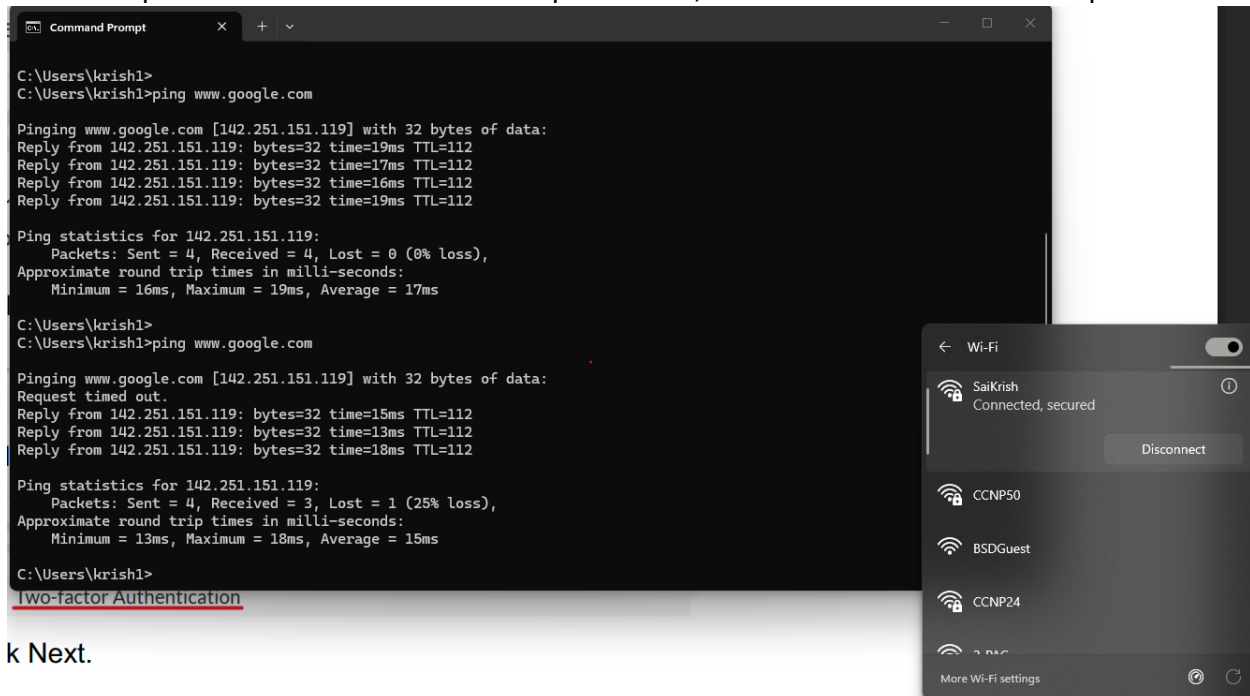
configure a firewall policy to allow traffic from the Enterprise WLAN to reach the WAN



Configure a name. Set the incoming/outgoing interface to the Enterprise WLAN/WAN interfaces respectively. Set the source/destination to the address object for the WLAN and “all” respectively. Set the service to ALL and the action to ACCEPT.

 This screenshot shows the 'Create New Policy' configuration window. The 'Name' field is set to 'energy'. The 'Schedule' is set to 'always'. The 'Action' is set to 'ACCEPT'. The 'Incoming interface' is set to 'SaiKrish (SaiKrish)' and the 'Outgoing interface' is set to 'Internet (wan)'. Under 'Source & Destination', the 'Source' is 'SaiKrish address' and the 'Destination' is 'all'. The 'Service' is set to 'ALL'. In the 'Firewall/Network Options' section, 'NAT' is enabled, and 'IP pool configuration' is set to 'Use Outgoing Interface Address'. Under 'Security Profiles', 'AntiVirus', 'Web filter', 'DNS filter', 'Application control', and 'IPS' are all disabled. 'SSL inspection' is set to 'no-inspection'. In the 'Logging Options' section, 'Log allowed traffic' is enabled, and 'Security events' is selected. A red arrow points to the 'OK' button at the bottom. On the right side, there is an 'Additional Information' panel with links to 'API Preview', 'Online Guides', 'Relevant Documentation', 'Video Tutorials', 'Consolidated Policy Configuration', 'Fortinet Community', and 'Join the Discussion'. At the bottom right, there is a 'Firmware Upgrade (1 device(s))' notification.

Your Enterprise WLAN has now been set up. To test it, connect from another WiFicapable device



k Next.

Problems

Our initial AP was a older AP and the OS on our Firewall gui, 7.6 didn't support that AP. We couldn't recognize the AP on that firewall, so we updraded our AP to the 421E and it was compatible to our firewall.

Conclusion

Overall this lab served as a great learning experience of setting up internet networks through Fortinet firewall and FortiAP. It helped me learn more about WLANS, WPA as well as The Fortinet ecosystem.